

Phase Noise and Timing Jitter in Oscillators With Colored-Noise Sources

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Abstract—Phase noise or timing jitter in oscillators is of major concern in wireless and optical communications, being a major contributor to the bit-error rate of communication systems, and creating synchronization problems in other clocked and sampled-data systems. This paper presents the theory and practical characterization of phase noise in oscillators due to *colored*, as opposed to white, noise sources. Shot and thermal noise sources in oscillators can be modeled as white-noise sources for all practical purposes. The characterization of phase noise in oscillators due to shot and thermal noise sources is covered by a recently developed theory of phase noise due to white-noise sources. The extension of this theory and the practical characterization techniques to noise sources in oscillators, which have a colored spectral density, e.g., $1/f$ noise, is crucial for practical applications. In this paper, we first derive a stochastic characterization of phase noise in oscillators due to colored-noise sources. This stochastic analysis is based on a novel *nonlinear perturbation analysis* for autonomous systems, and a *nonlocal Fokker–Planck equation* we derive. Then, we calculate the resulting spectrum of the oscillator output with phase noise as characterized. We also extend our results to the case when both white and colored-noise sources are present. Our treatment of phase noise due to colored-noise sources is general, i.e., it is not specific to a particular type of colored-noise source.

Index Terms—Circuit simulation, colored-noise sources, nonlinear oscillators, nonlocal Fokker–Planck theory, oscillator noise, phase noise, stochastic differential equations, timing jitter.

I. INTRODUCTION

OSCILLATORS are ubiquitous in physical systems, especially electronic and optical ones. For example, in radio frequency (RF) communication systems, they are used for frequency translation of information signals and for channel selection. Oscillators are also present in digital electronic systems which require a time reference, i.e., a clock signal, in order to synchronize operations.

Noise is of major concern in oscillators, because introducing even small noise into an oscillator leads to dramatic changes in its frequency spectrum and timing properties. This phenomenon, peculiar to oscillators, is known as *phase noise* or *timing jitter*. A perfect oscillator would have localized tones at discrete frequencies (i.e., harmonics), but any corrupting noise spreads these perfect tones, resulting in high power levels at neighboring frequencies. This effect is the major contributor to undesired phenomena such as interchannel interference, leading to increased bit-error rates in RF communication systems. Another manifestation of the same phenomenon, jitter, is important in clocked and sampled-data systems: uncertainties

in switching instants caused by noise lead to synchronization problems. Characterizing how noise affects oscillators is therefore crucial for practical applications. The problem is challenging, since oscillators constitute a special class among noisy physical systems: their *autonomous* nature makes them unique in their response to perturbations. A brief review of previous work on phase noise is given in [1].

In recent publications [1] and [2], we presented a theory and numerical methods for practical characterization of phase noise in oscillators with white-noise sources. In this paper, we present the theory and practical characterization of phase noise in oscillators due to colored, as opposed to white, noise sources. Shot and thermal noise sources in IC devices can be modeled as white-noise sources for all practical purposes. The characterization of phase noise in oscillators due to shot and thermal noise sources is covered by our theory of phase noise [1] due to white-noise sources. Other types of noise sources in IC devices, which have a colored spectral density (e.g., $1/f$ and burst noise), are significant in practical applications for the phase noise performance characterization of oscillators. We first derive the stochastic characterization of phase noise in oscillators due to colored-noise sources. Then, we calculate the resulting spectrum of an oscillator with phase noise characterized as above. We also extend our results to the case when both white and colored-noise sources are present. Our treatment of phase noise due to colored-noise sources is general, i.e., it is not specific to a particular type of colored-noise source. Hence, our results are applicable to the characterization of phase noise due to not only $1/f$ and burst noise, but also other types of possibly colored noise, e.g., substrate or power-supply noise.

The theory and numerical methods for phase noise are based on a novel nonlinear perturbation analysis that is valid for oscillators, which we summarize in Section II.¹ We review the characterization of phase noise due to white-noise sources in Section III. In Section IV, we analyze the case of colored-noise perturbations and obtain a stochastic characterization of the phase deviation. Models for burst (popcorn) and $1/f$ (flicker) noise, the most significant colored-noise sources in IC devices, are discussed in Section V. Then, in Section VI, we calculate the resulting oscillator spectrum with phase noise as characterized in Section IV. In Section VII, we consider the presence of white and colored-noise sources together, and derive the resulting oscillator spectrum. Finally, in Section VIII, we present simulation results in phase noise characterization of oscillators.

¹Originally in [1], we developed the nonlinear perturbation analysis for oscillators based on an autonomous *ordinary differential equation* (ODE) formulation. In [2], we generalized the nonlinear perturbation analysis theory to an autonomous index-1 *differential-algebraic equation* (DAE) formulation. Here, we summarize the perturbation analysis using an ODE formulation for notational simplicity.

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II. NONLINEAR PERTURBATION ANALYSIS FOR OSCILLATORS

In general, the dynamics of an oscillator can be described by a system of differential equations

$$\dot{x} = f(x) \quad (1)$$

where $x \in \mathbb{R}^n$ and $f(\cdot): \mathbb{R}^n \rightarrow \mathbb{R}^n$. We consider systems that have a periodic solution $x_s(t)$ (with period T) to (1), i.e., a stable-limit cycle in the n -dimensional solution space. We are interested in the response of such systems to a small state-dependent perturbation of the form $B(x)b(t)$ where $B(\cdot): \mathbb{R}^n \rightarrow \mathbb{R}^{n \times p}$ and $b(\cdot): \mathbb{R} \rightarrow \mathbb{R}^p$. Hence, the perturbed system is described by

$$\dot{x} = f(x) + B(x)b(t). \quad (2)$$

Although our eventual intent is to understand the response of the oscillator when $b(t)$ is random noise, it is useful to consider first the case when $b(t)$ is a known deterministic signal. We carried out a rigorous analysis of this case in [1] and obtained the following results.

- 1) The unperturbed oscillator's periodic response $x_s(t)$ is modified to $x_s(t + \alpha(t)) + y(t)$ by the perturbation, where
 - a) $\alpha(t)$ is a changing time shift, or *phase deviation*, in the periodic output of the unperturbed oscillator;
 - b) $y(t)$ is an additive component, which we term the *orbital deviation*, to the phase-shifted oscillator waveform.
- 2) $\alpha(t)$ and $y(t)$ are such that
 - a) $\alpha(t)$ will, in general, keep increasing with time even if the perturbation $b(t)$ is always small, and if the perturbation is removed, $\alpha(t)$ will settle to a constant value;
 - b) the orbital deviation $y(t)$, on the other hand, will always remain small [within a bounded factor of $b(t)$], and if the perturbation is removed, $y(t)$ will decay to zero.

Furthermore, we derived equations for $\alpha(t)$ and $y(t)$, in particular, a *nonlinear differential equation* for the phase deviation $\alpha(t)$:

$$\frac{d\alpha(t)}{dt} = v_1^T(t + \alpha(t))B(x_s(t + \alpha(t)))b(t), \quad \alpha(0) = 0 \quad (3)$$

where $v_1(t)$ is a periodically time-varying vector, which we call the *Floquet vector* [1], [2]. We will not go into a detailed derivation of (3) and the description of the Floquet vector here, which is covered elsewhere. In short, the Floquet vector $v_1(t)$ is obtained as follows.

- 1) The nonlinear differential equations (1) that describe the oscillator are linearized around the periodic steady-state solution $x_s(t)$ to obtain a *homogeneous* system of linear periodically time-varying (LPTV) differential equations, which has a periodic solution, the time derivative of $x_s(t)$, i.e., $\dot{x}_s(t)$.
- 2) The *adjoint* homogeneous LPTV system [2] is formed, which also has a periodic solution. Any scaled version of the periodic solution of the adjoint system is also a solution for it. The Floquet vector $v_1(t)$ is obtained by scaling this periodic solution so that it satisfies $v_1^T(t)\dot{x}_s(t) = 1$.

The Floquet vector plays a crucial role in our analysis. For the derivations and proofs of the above results, we use the Floquet theory [3] of LPTV ODE systems, and its generalization to index-1 DAE systems [2]. We developed numerical methods [1], [2], both in time and frequency domain, for the efficient computation of the periodic Floquet vector $v_1(t)$ in (3), which is sufficient to characterize both spectral spreading and timing jitter in oscillators.

From (3), $\alpha(t)$ grows very much like the integral of the perturbation. For a constant perturbation, for example, $\alpha(t)$ is approximately a linear ramp. This indicates how the *frequency of the oscillator can change due to perturbations*, for a linearly increasing phase error is equivalent to a frequency error. In [1], we concretized the above observation for white-noise perturbations, which we summarize in Section III.

III. PHASE NOISE IN OSCILLATORS WITH WHITE-NOISE SOURCES

In [1], we considered the case where the perturbation $b(t)$ is a vector of (uncorrelated) stationary, white Gaussian noise processes—this situation is important for determining practical figures of merit like zero-crossing jitter and spectral purity (i.e., spreading of the power spectrum) when the thermal and shot noise sources in IC devices are considered, which can be modeled as modulated stationary white-noise processes. The state-dependent modulation is in $B(x_s(t))$. For example, the shot noise intensity in a *pn*-junction depends on the (periodic) large-signal current through the junction. Jitter and spectral spreading are closely related, and both are determined by the manner in which $\alpha(t)$, now also a random process, spreads with time. In [1], for white-noise excitations, we established the following.

- 1) The average spread of the jitter (mean-square jitter, or variance) increases precisely *linearly* with time, i.e.

$$\mathbb{E}[\alpha^2(t)] = \sigma^2(t) = ct \quad (4)$$

where

$$c = \frac{1}{T} \int_0^T v_1^T(\tau)B(x_s(\tau))B^T(x_s(\tau))v_1(\tau) d\tau. \quad (5)$$

- 2) The power spectrum of the perturbed oscillator is a *Lorentzian* about each harmonic. A Lorentzian is the shape of the squared magnitude of a one-pole lowpass filter transfer function.
- 3) A *single scalar constant* c is sufficient to describe jitter and spectral spreading in a noisy oscillator with white-noise sources.
- 4) The oscillator's output with phase noise due to white-noise sources, i.e., $x_s(t + \alpha(t))$ is a *stationary* stochastic process.

If we define X_i to be the Fourier coefficients of $x_s(t)$

$$x_s(t) = \sum_{i=-\infty}^{\infty} X_i \exp(ji2\pi f_0 t)$$

then, the spectrum of the stationary oscillator output $x_s(t + \alpha(t))$ with white-noise sources is given by

$$S_V(f) = \sum_{i=-\infty}^{\infty} X_i X_i^* \frac{f_0^2 i^2 c}{\pi^2 f_0^4 i^4 c^2 + (f + if_0)^2} \quad (6)$$

where $f_0 = 1/T$ is the fundamental frequency. The phase deviation $\alpha(t)$ does *not* change the total power in the periodic signal $x_s(t)$, but it alters the power density in frequency, i.e., the power spectral density. For the perfect periodic signal $x_s(t)$, the power spectral density has δ functions located at discrete frequencies (i.e., the harmonics). The phase deviation $\alpha(t)$ *spreads* the power in these δ functions in the form given in (6), which can be experimentally observed with a spectrum analyzer.

The above results have important implications. The power spectral density at the carrier frequency and its harmonics has a finite value, and that the total carrier power is preserved despite spectral spreading due to noise. Previous analyzes based on linear time-invariant or linear time-varying concepts erroneously predict infinite noise power density at the carrier, as well as infinite total integrated power. That the oscillator output is stationary is surprising at first sight, since oscillators are nonlinear systems with periodic swings, hence it might be expected that output noise power would change periodically as in forced systems. However, it must be remembered that while forced systems are supplied with an external time reference (through the forcing), oscillators are not. Cyclostationarity in the oscillator's output would, by definition, imply a time reference. Hence, the stationarity result reflects the fundamental fact that noisy autonomous systems cannot provide a perfect time reference.

IV. STOCHASTIC CHARACTERIZATION OF THE PHASE DEVIATION WITH COLORED NOISE SOURCES

We now find the probabilistic characterization of the phase deviation $\alpha(t)$ [which satisfies the differential equation (3)] as a stochastic process when the perturbation $b(t)$ is a (one-dimensional) stationary, zero-mean ($\mathbb{E}[b(t)] = 0$), Gaussian colored stochastic process.² Let $R_N(\tau)$ be the autocovariance function, and $S_N(f)$ be the power spectral density, of the stationary Gaussian stochastic process $b(t)$

$$R_N(\tau) = \mathbb{E}[b(t + \tau/2)b(t - \tau/2)] \quad (7)$$

$$S_N(f) = \mathbb{F}\{R_N(\tau)\} = \int_{-\infty}^{\infty} R_N(\tau) \exp(-j2\pi f\tau) d\tau. \quad (8)$$

Note that $R_N(\tau)$ is a real and even function of τ . Let

$$v(t) = v_1^T(t)B(x_s(t)) \quad (9)$$

which is a scalar [both $v_1(\cdot)$ and $B(\cdot)$ are vectors] that is periodic in t with period T . Hence, (3) becomes

$$\frac{d\alpha(t)}{dt} = v(t + \alpha(t))b(t), \quad \alpha(0) = 0. \quad (10)$$

In this section, we will follow the below procedure to find an adequate probabilistic characterization of the phase deviation $\alpha(t)$ due to the colored-noise source $b(t)$ for our purposes.

²The extension to the case when $b(t)$ is a vector of uncorrelated white and colored-noise sources is discussed in Section VII. Noise sources in electronic devices usually have independent physical origin, and hence they are modeled as uncorrelated stochastic processes. Hence, we consider uncorrelated noise sources. However, the generalization of our results to correlated noise sources is trivial.

- 1) We first derive a *partial integro-differential equation* for the time-varying marginal *probability density function* (PDF) $p_\alpha(\eta, t)$ of $\alpha(t)$ defined as

$$p_\alpha(\eta, t) = \frac{\partial \mathbb{P}(\alpha(t) \leq \eta)}{\partial \eta}, \quad t \geq 0 \quad (11)$$

where $\mathbb{P}(\cdot)$ denotes the *probability measure*.

- 2) We then show that the PDF of a Gaussian random variable, “asymptotically” with t , solves this partial integro-differential equation. A Gaussian PDF is completely characterized by the mean and the variance. We show that $\alpha(t)$ becomes [under some conditions on $R_N(\tau)$ or $S_N(f)$], for “large” (to be concretized) t , a Gaussian random variable with a constant mean and a variance that is given by

$$\text{var}(\alpha(t)) = \beta \int_0^t \int_0^t R_N(t_1 - t_2) dt_1 dt_2. \quad (12)$$

The quantity in (12) (excluding the constant factor β) is the variance of the integral of $b(t)$, i.e.

$$\mathbb{E} \left[\left(\int_0^t b(s) ds \right)^2 \right] = \int_0^t \int_0^t R_N(t_1 - t_2) dt_1 dt_2. \quad (13)$$

Theorem IV.1: If $b(t)$ is a stationary, zero-mean, Gaussian stochastic process with autocovariance function $R_N(\tau)$, and if $\alpha(t)$ satisfies (10), then the time-varying marginal PDF $p_\alpha(\eta, t)$ of α satisfies

$$\begin{aligned} \frac{\partial p_\alpha(\eta, t)}{\partial t} &= -\frac{\partial}{\partial \eta} \left(\frac{\partial v(t + \eta)}{\partial \eta} \int_0^t v(\tau + \eta) R_N(t - \tau) p_\alpha(\eta, t) d\tau \right) \\ &\quad + \frac{\partial^2}{\partial \eta^2} \left(v(t + \eta) \int_0^t v(\tau + \eta) R_N(t - \tau) p_\alpha(\eta, t) d\tau \right) \end{aligned} \quad (14)$$

with the initial/boundary condition for $t = 0$

$$p_\alpha(\eta, 0) = \delta(\eta) \quad (15)$$

i.e.

$$\begin{aligned} \mathbb{E}[\alpha(0)] &= 0 \\ \mathbb{E}[\alpha^2(0)] &= 0. \end{aligned}$$

Proof: See the Appendix. ■

The partial integro-differential equation (14) for the time-varying marginal PDF $p_\alpha(\eta, t)$ of $\alpha(t)$ is a generalization of a partial differential equation known as the *Fokker-Planck equation* [4], [5] derived for the PDF of $\alpha(t)$ satisfying (10) when $b(t)$ is a white-noise process, which is given as

$$\begin{aligned} \frac{\partial p_\alpha(\eta, t)}{\partial t} &= -\frac{\partial}{\partial \eta} \left(\lambda \frac{\partial v(t + \eta)}{\partial \eta} v(t + \eta) p_\alpha(\eta, t) \right) \\ &\quad + \frac{1}{2} \frac{\partial^2}{\partial \eta^2} (v^2(t + \eta) p_\alpha(\eta, t)) \end{aligned} \quad (16)$$

where $0 \leq \lambda \leq 1$ depends on the definition of the stochastic integral [4] used to interpret the stochastic differential equation in (10) with $b(t)$ as a white-noise process. If $b(t)$ is a white-noise process, then $\alpha(t)$ is a Markov process. However, when $b(t)$ is colored, $\alpha(t)$, in general, is not Markovian. Equation (16) is

valid for any initial/boundary condition. On the other hand, (14) is valid only for initial/boundary conditions of the type

$$p_\alpha(\eta, 0) = \delta(\eta - \alpha_0) \quad (17)$$

for some α_0 .

We would like to solve (14) for $p_\alpha(\eta, t)$. We do this by first solving for the *characteristic function* $F(\omega, t)$ of $\alpha(t)$, which is defined by

$$F(\omega, t) = \mathbb{E}[\exp(j\omega\alpha(t))] = \int_{-\infty}^{\infty} \exp(j\omega\eta) p_\alpha(\eta, t) d\eta. \quad (18)$$

$v(t)$ is T -periodic, hence we can expand $v(t)$ into its Fourier series

$$v(t) = \sum_{i=-\infty}^{\infty} V_i \exp(ji\omega_0 t), \quad \omega_0 = \frac{2\pi}{T}.$$

Lemma IV.1: The characteristic function of $\alpha(t)$, $F(\omega, t)$, satisfies

$$\begin{aligned} \frac{\partial F(\omega, t)}{\partial t} &= \sum_{i=-\infty}^{\infty} \sum_{k=-\infty}^{\infty} V_i V_k^* \exp(j\omega_0 i t) (-\omega_0 i \omega - \omega^2) \\ &\cdot \int_0^t R_N(t - \tau) \exp(-j\omega_0 k \tau) F(\omega_0(i - k) + \omega, t) d\tau \end{aligned} \quad (19)$$

where $*$ denotes complex conjugation.

Proof: Equation (19) is obtained from (14) by using the definition of the characteristic function in (18). ■

Theorem IV.2: Equation (19) has a solution that becomes (with time) the characteristic function of a Gaussian random variable

$$F(\omega, t) = \exp\left(j\omega\mu(t) - \frac{\omega^2\sigma^2(t)}{2}\right) \quad (20)$$

solves (19) for t large enough such that

$$\exp\left(-\frac{1}{2}\omega_0^2(i - k)^2\sigma^2(t)\right) \approx \begin{cases} 1, & i = k \\ 0, & i \neq k \end{cases} \quad (21)$$

where

$$\frac{d\mu(t)}{dt} = \sum_{i=-\infty}^{\infty} j\omega_0 i |V_i|^2 \int_0^t R_N(t - \tau) \exp(j\omega_0 i(t - \tau)) d\tau \quad (22)$$

$$\frac{d\sigma^2(t)}{dt} = \sum_{i=-\infty}^{\infty} 2|V_i|^2 \int_0^t R_N(t - \tau) \exp(j\omega_0 i(t - \tau)) d\tau. \quad (23)$$

Assumption IV.1:

$$\begin{aligned} &\int_0^t R_N(t - \tau) \exp(j\omega_0 i(t - \tau)) d\tau \\ &\approx \begin{cases} \int_0^t R_N(t - \tau) d\tau, & i = 0 \\ 0, & i \neq 0. \end{cases} \end{aligned} \quad (24)$$

This is satisfied when the *bandwidth* of the colored-noise source is much less than the oscillation frequency ω_0 , or equivalently, the *correlation width* of the colored-noise source in time is much

larger than the oscillation period $T = 2\pi/\omega_0$. We will further comment on this condition in Section V, where we discuss the models for burst and $1/f$ noise. With (24), (22) and (23) become

$$\frac{d\mu(t)}{dt} = 0 \quad (25)$$

$$\frac{d\sigma^2(t)}{dt} = 2|V_0|^2 \int_0^t R_N(t - \tau) d\tau. \quad (26)$$

From (26)

$$\sigma^2(t) = 2|V_0|^2 \int_0^t \int_0^t R_N(t_2 - t_1) dt_1 dt_2 \quad (27)$$

follows trivially. Since, the autocovariance $R_N(\tau)$ is an even function of τ , (27) can be rewritten as

$$\sigma^2(t) = |V_0|^2 \int_0^t \int_0^t R_N(t_2 - t_1) dt_1 dt_2. \quad (28)$$

Thus, we obtained (12).

Lemma IV.2: The variance $\sigma^2(t)$ of $\alpha(t)$ in (27) can be rewritten with a single integral as follows:

$$\sigma^2(t) = 2|V_0|^2 \int_0^t (t - \tau) R_N(\tau) d\tau. \quad (29)$$

It can also be expressed in terms of the spectral density of the colored-noise source $b(t)$ as follows:

$$\sigma^2(t) = 2|V_0|^2 \int_{-\infty}^{\infty} S_N(f) \frac{(1 - \exp(j2\pi ft))}{4\pi^2 f^2} df. \quad (30)$$

V. MODELS FOR BURST (POPCORN) AND $1/f$ (FLICKER) NOISE

A. Burst Noise

The source of burst noise is not fully understood, although it has been shown to be related to the presence of heavy-metal ion contamination [6]. Gold-doped devices show very high levels of burst noise. For practical purposes, burst noise is usually modeled with a colored stochastic process with Lorentzian spectrum, i.e., the spectral density of a burst noise source is given by

$$S_{\text{burst}}(f) = K \frac{I^a}{1 + \left(\frac{f}{f_c}\right)^2} \quad (31)$$

where K is a constant for a particular device, I is the current through the device, a is a constant in the range 0.5 to 2, and f_c is the 3-dB bandwidth of the Lorentzian spectrum [6]. Burst noise often occurs with multiple time constants, i.e., the spectral density is the summation of several Lorentzian spectra as given by (31) with different 3-dB bandwidths.

A stationary colored stochastic process with spectral density

$$S_{N\text{burst}}(f) = \frac{\gamma^2}{\gamma^2 + (2\pi f)^2} \quad (32)$$

has the autocorrelation function

$$R_{N\text{burst}}(\tau) = \frac{\gamma}{2} \exp(-\gamma|\tau|). \quad (33)$$

If the 3-dB bandwidth γ of (32) is much less than the oscillation frequency ω_0 , or equivalently, the correlation width $1/\gamma$ of (33) is much larger than the oscillation period $T = 2\pi/\omega_0$, then (24) is satisfied.

The variance of the integral of a stationary stochastic process $b(t)$ with spectral density as in (32) is given by

$$\mathbb{E} \left[\left(\int_0^t b(s) ds \right)^2 \right] = \int_0^t \int_0^t R_{N\text{burst}}(t_1 - t_2) dt_1 dt_2 \quad (34)$$

$$= \frac{\gamma t - 1 + \exp(-\gamma t)}{\gamma}. \quad (35)$$

B. $1/f$ Noise

$1/f$ noise is ubiquitous in all physical systems (as a matter of fact, in all kinds of systems). The origins of $1/f$ noise is varied. In IC devices, it is believed to be caused mainly by traps associated with contamination and crystal defects, which capture and release charge carriers in a random fashion, and the time constants associated with this process give rise to a noise signal with energy concentrated at low frequencies. For practical purposes it is modeled with a “stationary” and colored stochastic process with a spectral density given by

$$S_{1/f}(f) = K \frac{I^a}{f} \quad (36)$$

where K is a constant for a particular device, I is the current through the device, and a is a constant in the range 0.5 to 2. There is a lot of controversy both about the origins and modeling of $1/f$ noise. The spectral density in (36) is not a well-defined spectral density for a stationary stochastic process. It blows up at $f = 0$. Keshner in [7] argues that $1/f$ noise is *really* a non-stationary process, and when one tries to model it as a stationary process, this nonphysical artifact arises. We are not going to dwell into this further, which would fill up pages and would not be too useful other than creating a lot of confusion. Instead, we will “postulate” a well-defined *stationary* stochastic process model for $1/f$ noise: We will introduce a cutoff frequency in (36), below which the spectrum deviates from $1/f$ and attains a finite value at $f = 0$. To do this, we use the following integral representation [8]

$$\frac{1}{|f|} = 4 \int_0^\infty \frac{1}{\gamma^2 + (2\pi f)^2} d\gamma. \quad (37)$$

We introduce the cutoff frequency γ_c in (37), and use

$$S_{N1/f}(f) = 4 \int_{\gamma_c}^\infty \frac{1}{\gamma^2 + (2\pi f)^2} d\gamma \quad (38)$$

$$= \frac{1}{|f|} - 4 \frac{\arctan\left(\frac{\gamma_c}{2\pi f}\right)}{2\pi f} \quad (39)$$

for the spectral density of a stationary stochastic process that models $1/f$ noise. The spectral density in (38) has a finite value at $f = 0$

$$S_{N1/f}(0) = \frac{4}{\gamma_c}. \quad (40)$$

The autocorrelation function that corresponds to the spectral density in (38) is given by

$$R_{N1/f}(\tau) = 2E_1(\gamma_c|\tau|) \quad (41)$$

where the exponential integral $E_1(z)$ is defined as

$$E_1(t) = \int_1^\infty \frac{\exp(-tz)}{z} dz.$$

The power in a $1/f$ noise source modeled with a stochastic process with the spectral density (38) is concentrated at low frequencies, frequencies much less than the oscillation frequency for practical oscillators. Hence, (24) is satisfied for $1/f$ noise sources.

The variance of the integral of a stationary stochastic process $b(t)$ with spectral density as in (38) is given by

$$\begin{aligned} \mathbb{E} \left[\left(\int_0^t b(s) ds \right)^2 \right] &= \int_0^t \int_0^t R_{N1/f}(t_1 - t_2) dt_1 dt_2 \\ &= 2 \frac{2\gamma_c t - 1 + \exp(-\gamma_c t) - \gamma_c t \exp(-\gamma_c t) + (\gamma_c t)^2 E_1(\gamma_c t)}{\gamma_c^2}. \end{aligned} \quad (42)$$

VI. SPECTRUM OF AN OSCILLATOR WITH PHASE NOISE DUE TO COLORED NOISE SOURCES

Having obtained the stochastic characterization of $\alpha(t)$ due to a colored-noise source in Section IV, we now compute the spectral density of the oscillator output, i.e., $x_s(t + \alpha(t))$. We first obtain an expression for the nonstationary autocovariance function $R(t, \tau)$ of $x_s(t + \alpha(t))$. Next, we demonstrate that the autocovariance is independent of t for “large” time. Finally, we calculate the spectral density of $x_s(t + \alpha(t))$ by taking the Fourier transform of the stationary autocovariance function for $x_s(t + \alpha(t))$.

We start by calculating the autocovariance function of $x_s(t + \alpha(t))$, given by

$$R_V(t, \tau) = \mathbb{E}[x_s(t + \alpha(t))x_s^*(t + \tau + \alpha(t + \tau))]. \quad (43)$$

Definition VI.1: Define X_i to be the Fourier coefficients of $x_s(t)$

$$x_s(t) = \sum_{i=-\infty}^{\infty} X_i \exp(ji\omega_0 t).$$

The following simple Lemma establishes the basic form of the autocovariance:

Lemma VI.1:

$$\begin{aligned} R_V(t, \tau) &= \sum_{i=-\infty}^{\infty} \sum_{k=-\infty}^{\infty} X_i X_k^* \exp(j(i - k)\omega_0 t) \\ &\quad \cdot \exp(-jk\omega_0 \tau) \mathbb{E}[\exp(j\omega_0(i\alpha(t) - k\alpha(t + \tau)))]. \end{aligned} \quad (44)$$

The expectation in (44), i.e., $\mathbb{E}[\exp(j\omega_0(i\alpha(t) - k\alpha(t + \tau)))]$ is the characteristic function of $i\alpha(t) - k\alpha(t + \tau)$. This expectation is independent of t for large time as established by the following theorem:

Theorem VI.1: If t is large enough such that

$$\exp\left(-\frac{1}{2}\omega_0^2(i - k)^2\sigma^2(t)\right) \approx \begin{cases} 1, & i = k \\ 0, & i \neq k \end{cases} \quad (45)$$

then $i\alpha(t) - k\alpha(t + \tau)$ is a Gaussian random variable and its characteristic function, which is independent of t , is given by

$$\begin{aligned} E[\exp(j\omega_0(i\alpha(t) - k\alpha(t + \tau)))] \\ \approx \begin{cases} 0, & i \neq k \\ \exp(-\frac{1}{2}\omega_0^2 i^2 \sigma^2(|\tau|)), & i = k \end{cases} \end{aligned} \quad (46)$$

where $\sigma^2(t)$ is as in (28), (29) or (30). Note that the condition (45) is same as the condition (21) in Theorem IV.2.

We now obtain the stationary autocovariance function from (44) using (46).

Corollary VI.1:

$$R_V(\tau) = \sum_{i=-\infty}^{\infty} X_i X_i^* \exp(-ji\omega_0\tau) \exp(-\frac{1}{2}\omega_0^2 i^2 \sigma^2(|\tau|)). \quad (47)$$

To obtain the spectral density of $x_s(t + \alpha(t))$, we calculate the Fourier transform of (47):

$$S_V(f) = \sum_{i=-\infty}^{\infty} X_i X_i^* S_i(f + if_0) \quad (48)$$

where $\omega_0 = 2\pi f_0$ and

$$S_i(f) = F\{R_i(\tau)\} \quad (49)$$

$$= F\left\{\exp\left(-\frac{1}{2}\omega_0^2 i^2 \sigma^2(|\tau|)\right)\right\}. \quad (50)$$

We reproduce $\sigma^2(t)$ in (27)–(30) here for convenience

$$\sigma^2(t) = 2|V_0|^2 \int_0^t \int_0^{t_2} R_N(t_2 - t_1) dt_1 dt_2 \quad (51)$$

$$= |V_0|^2 \int_0^t \int_0^t R_N(t_2 - t_1) dt_1 dt_2 \quad (52)$$

$$= 2|V_0|^2 \int_0^t (t - \tau) R_N(\tau) d\tau \quad (53)$$

$$= 2|V_0|^2 \int_{-\infty}^{\infty} S_N(f) \frac{(1 - \exp(j2\pi ft))}{4\pi^2 f^2} df \quad (54)$$

where

$$S_N(f) = F\{R_N(\tau)\}$$

is the autocovariance/spectral density pair for the colored-noise source $b(t)$.

The Fourier transform in (50) does not have a simple closed form. Mullen and Middleton in [9] calculate various limiting forms for this Fourier transform through approximating series expansions. We are going to use some of their methods to calculate limiting forms for (50) for different frequency ranges of interest, but before that, we would like to establish some basic, general properties for $\sigma^2(t)$, $R_i(\tau)$ and $S_i(f)$.

Lemma VI.2:

$$\lim_{t \rightarrow \infty} \frac{\sigma^2(t)}{t} = |V_0|^2 S_N(0). \quad (55)$$

Proof: Equation (55) is obtained using (51). ■

Corollary VI.2:

$$\lim_{t \rightarrow \infty} \sigma^2(t) = \begin{cases} \infty, & S_N(0) \neq 0 \\ \sigma^2(\infty), & S_N(0) = 0 \end{cases} \quad (56)$$

where $\sigma^2(\infty) < \infty$ is a finite nonnegative value.

For the models of burst and $1/f$ noise discussed in Section V, we have

$$S_{N\text{burst}}(0) = 1$$

from (32), and

$$S_{N1/f}(0) = \frac{4}{\gamma_c}$$

from (40). Thus, $S_N(0) \neq 0$, and hence, $\lim_{t \rightarrow \infty} \sigma^2(t) = \infty$ are satisfied for both.

Since $\sigma^2(0) = 0$ we have

$$R_i(0) = 1 \quad (57)$$

and hence

$$\int_{-\infty}^{\infty} S_i(f) df = 1 \quad (58)$$

for any colored-noise source. The total power $X_i X_i^*$ in the i th harmonic of the spectrum is preserved. The distribution of the power in frequency is given by $S_i(f)$.

Lemma VI.3: If

$$\lim_{t \rightarrow \infty} \sigma^2(t) = \infty \quad (59)$$

then $S_i(0)$ is nonnegative and finite. If

$$\lim_{t \rightarrow \infty} \sigma^2(t) = \sigma^2(\infty) < \infty \quad (60)$$

then

$$S_i(f) = \exp\left(-\frac{1}{2}\omega_0^2 i^2 \sigma^2(\infty)\right) \delta(f) + \tilde{S}_i(f)$$

where $\tilde{S}_i(0)$ is nonnegative and finite, and

$$\int_{-\infty}^{\infty} \tilde{S}_i(f) df = 1 - \exp\left(-\frac{1}{2}\omega_0^2 i^2 \sigma^2(\infty)\right).$$

Now, we concentrate on the case when (59) is satisfied, i.e., when the spectrum takes a finite value at the carrier frequency (and its harmonics). Next, we proceed as Mullen and Middleton in [9] and calculate limiting forms to the Fourier transform in (50) through approximating series:

Theorem VI.2: Let (59) be true. For f away from 0, (50) can be approximated with

$$S_i(f) \approx i^2 |V_0|^2 \frac{f_0^2}{f^2} S_N(f), \quad f \gg 0 \quad (61)$$

where $\omega_0 = 2\pi f_0$. For f around 0, (50) can be approximated with

$$\begin{aligned} S_i(f) \approx & F\left\{\exp\left(-\frac{1}{2}\omega_0^2 i^2 |V_0|^2 S_N(0) |\tau|\right)\right\} \\ & + F\left\{\exp\left(-\frac{1}{2}\omega_0^2 i^2 |V_0|^2 S_N(0) |\tau|\right)\right\} \\ & \otimes F\left\{\omega_0^2 i^2 |V_0|^2 \left(|\tau| \int_{|\tau|}^{\infty} R_N(z) dz \right.\right. \\ & \left.\left. + \int_0^{|\tau|} z R_N(z) dz\right)\right\} \end{aligned}$$

$$\begin{aligned} &\approx \frac{f_0^2 i^2 (|V_0|^2 S_N(0))}{\pi^2 f_0^4 i^4 (|V_0|^2 S_N(0))^2 + f^2} \\ &+ \frac{f_0^2 i^2 (|V_0|^2 S_N(0))}{\pi^2 f_0^4 i^4 (|V_0|^2 S_N(0))^2 + f^2} \\ &\otimes F \left\{ \omega_0^2 i^2 |V_0|^2 \left(|\tau| \int_{|\tau|}^{\infty} R_N(z) dz \right. \right. \\ &\quad \left. \left. + \int_0^{|\tau|} z R_N(z) dz \right) \right\} \quad (62) \end{aligned}$$

where $\otimes$ denotes convolution. The first term in (62) is a Lorentzian with corner frequency

$$\pi f_0^2 i^2 (|V_0|^2 S_N(0))$$

and can be used as an approximation for (50) around $f = 0$ by ignoring the higher order second term. Equation (62) contains the first two terms of a series expansion for (50).

Proof: Equation (61) is obtained using the representation of $\sigma^2(t)$ in (54), and (62) is obtained using (53). ■

From (61), we observe that the frequency dependence of $S_i(f)$ is as $1/f^2$ multiplied with the spectral density $S_N(f)$ of the noise source. This result matches with measurement results for phase noise spectrum due to $1/f$ noise sources.

VII. PHASE NOISE AND SPECTRUM OF AN OSCILLATOR DUE TO WHITE AND COLORED-NOISE SOURCES

We now consider the case when both white and colored-noise sources are present. Let there be p white-noise sources and M colored-noise sources

$$\dot{x} = f(x) + B_w(x)b_w(t) + \sum_{m=1}^M B_{cm}(x)b_{cm}(t) \quad (63)$$

where $B_w(\cdot): \mathbb{R}^n \rightarrow \mathbb{R}^{n \times p}$, $B_{cm}(\cdot): \mathbb{R}^n \rightarrow \mathbb{R}^n$, $m = 1, \dots, M$, $b_w(\cdot): \mathbb{R} \rightarrow \mathbb{R}^p$ is a vector of (uncorrelated) stationary, normalized,³ white Gaussian noise processes, and $b_{cm}(\cdot): \mathbb{R} \rightarrow \mathbb{R}$, $m = 1, \dots, M$ are zero-mean, Gaussian, stationary colored stochastic processes [uncorrelated with each other, and with $b_w(t)$] with autocorrelation function/spectral density pairs

$$S_{Nm}(f) = F\{R_{Nm}(\tau)\}, \quad m = 1, \dots, M$$

where

$$S_{Nm}(0) \neq 0, \quad m = 1, \dots, M$$

and $R_{Nm}(\tau)$ $m = 1, \dots, M$ are assumed to satisfy (24). In this case, the phase error $\alpha(t)$ satisfies the nonlinear differential equation

$$\begin{aligned} \frac{d\alpha(t)}{dt} &= v_1^T(t + \alpha(t)) \\ &\cdot \left[B_w(x_s(t + \alpha(t)))b_w(t) + \sum_{m=1}^M B_{cm}(x_s(t + \alpha(t)))b_{cm}(t) \right] \quad (64) \end{aligned}$$

³The spectrum is normalized to 1 (for all frequency), and the amplitude for b_w is set by the multiplicative modulating factor B_w in (63).

where $v_1(t)$ is the periodically time-varying Floquet vector [1], [2] of Section II. Let

$$c_w = \frac{1}{T} \int_0^T v_1^T(\tau) B_w(x_s(\tau)) B_w^T(x_s(\tau)) v_1(\tau) d\tau \quad (65)$$

and

$$V_{0m} = \frac{1}{T} \int_0^T v_1^T(\tau) B_{cm}(x_s(\tau)) d\tau \quad m = 1, \dots, M. \quad (66)$$

Lemma VII.1: $\alpha(t)$ that satisfies (64) becomes a Gaussian random variable with constant mean, and variance given by

$$\begin{aligned} \sigma^2(t) &= c_w t + \sum_{m=1}^M 2|V_{0m}|^2 \int_0^t (t - \tau) R_{Nm}(\tau) d\tau \\ &= c_w t + \sum_{m=1}^M 2|V_{0m}|^2 \int_{-\infty}^{\infty} S_{Nm}(f) \frac{(1 - \exp(j2\pi ft))}{4\pi^2 f^2} df \end{aligned}$$

for t large enough such that

$$\exp\left(-\frac{1}{2}\omega_0^2(i - k)^2\sigma^2(t)\right) \approx \begin{cases} 1, & i = k \\ 0, & i \neq k. \end{cases} \quad (67)$$

Theorem VII.1: With $\alpha(t)$ characterized as above, the oscillator output $x_s(t + \alpha(t))$ is a stationary process, and its spectral density is given by

$$S_V(f) = \sum_{i=-\infty}^{\infty} X_i X_i^* S_i(f + if_0) \quad (68)$$

where X_i are the Fourier series coefficients of $x_s(t)$, and

$$S_i(f) = \begin{cases} \frac{f_0^2 i^2 \left(c_w + \sum_{m=1}^M |V_{0m}|^2 S_{Nm}(0) \right)}{\pi^2 f_0^4 i^4 \left(c_w + \sum_{m=1}^M |V_{0m}|^2 S_{Nm}(0) \right)^2 + f^2}, & f \approx 0 \\ i^2 \frac{f_0^2}{f^2} \left(c_w + \sum_{m=1}^M |V_{0m}|^2 S_{Nm}(f) \right), & f \gg 0. \end{cases} \quad (69)$$

The full spectrum of the oscillator with white and colored-noise sources has the shape of a Lorentzian around the carrier, and away from the carrier, the white-noise sources contribute a term that has a $1/f^2$ frequency dependence, and the colored-noise sources contribute terms that have a frequency dependence as $1/f^2$ multiplied with the spectral density of the colored-noise source.

VIII. EXAMPLES

We have derived an *analytical* expression, given by (68) and (69), for the spectrum of the oscillator output with phase noise due to white and colored-noise sources. The analytical expression in (68) and (69) contains some parameters to be computed.

- X_i : The Fourier series coefficients of the large-signal *noiseless* periodic waveform $x_s(t)$ of the oscillator output.
- $c_w = (1/T) \int_0^T v_1^T(\tau) B_w(x_s(\tau)) B_w^T(x_s(\tau)) v_1(\tau) d\tau$: Scalar that characterizes the contributions of the white-noise sources.

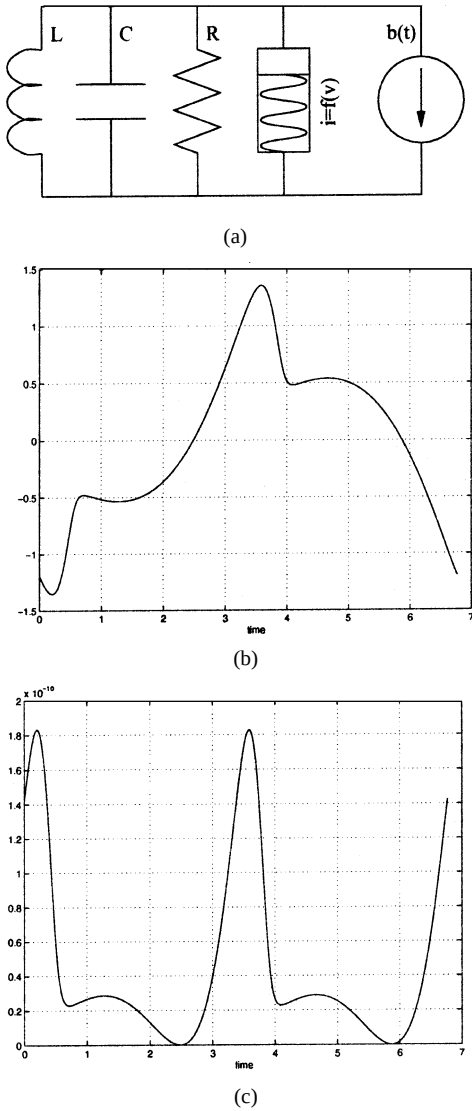


Fig. 1. Simple oscillator with parallel RLC-nonlinearity.

- $V_{0m} = (1/T) \int_0^T v_1^T(\tau) B_{cm}(x_s(\tau)) d\tau$ $m = 1, \dots, M$: Scalars that characterize the contributions of the colored-noise sources.

We have developed efficient numerical methods, which were implemented in a circuit simulator, for the computation of the periodic Floquet vector $v_1(t)$, and hence for the computation of the parameters above. Once the periodic steady state $x_s(t)$ of the oscillator and the scalars c_w and V_{0m} , $m = 1, \dots, M$ are computed, we have an analytical expression that gives us the spectrum of the oscillator at *any* frequency f . The computation of the spectrum is *not* performed separately for every frequency of interest. We compute the whole spectrum as a function of frequency *at once*, which makes our technique very efficient.

Moreover, as will be illustrated below, one can perform a detailed analysis of the mechanism of phase noise generation using our techniques. In particular, one can calculate the contribution of the noise sources separately.

Oscillator With Parallel RLC and a Nonlinear Current Source

We now present simulation results in the phase noise characterization of a simple oscillator in Fig. 1(a). The resistor is

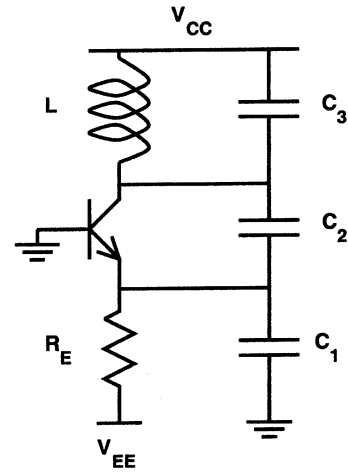


Fig. 2. Colpitts oscillator.

assumed to be noiseless, but we insert a stationary external current noise source across the capacitor. The Floquet vector $v_1(t)$ is a two-dimensional vector, since the oscillator has two state variables, namely the capacitor voltage and the inductor current. Fig. 1(b) shows the entry of $v_1(t)$ corresponding to the capacitor voltage.

Now, let us assume that we have two current noise sources across the capacitor, one of them a white stationary noise source, and the other a colored stationary noise source with bandwidth much smaller than the oscillation frequency. To calculate the spectrum of the capacitor voltage given by (68) and (69), we need to compute c_w in (65) for the white-noise source, and V_0 in (66) for the colored-noise source. For stationary noise sources, $B_w(t)$ in (65) and $B_c(t)$ in (66) are constant functions of time t . Fig. 1(c) shows $v_1^T(t) B_w B_w^T v_1(t)$ which is a periodic function of time t . Note that c_w is the time-average of this quantity.

From (66), we observe that V_0 is the time-average of $v_1^T(t) B_c$. The time-average of $v_1(t)$ in Fig. 1(b) for the capacitor voltage is 0! Thus, we conclude that any *stationary* [with the modulation $B_c(t)$ a constant function of time] colored-noise source (with bandwidth much smaller than the oscillation frequency) connected across the capacitor has no contribution to the oscillator spectrum due to phase noise, because $V_0 = 0$ for this noise source.

Colpitts Oscillator

The oscillator for this example is a standard Colpitts oscillator with a single bipolar transistor as the active device, shown in Fig. 2 ($V_{CC} = 5$, $V_{EE} = -5$, $L = 1$ nH, $R_{EE} = 5$ K Ω , $C_1 = 10$ pF, $C_2 = 4$ pF, $C_3 = 10$ pF). Fig. 3(a) and (b) show the difference of the entries of the Floquet vector $v_1(t)$ for the *base-emitter* nodes, and *collector-emitter* nodes of the transistor, respectively.

The waveform $v_1(t)$ for *collector-emitter* is multiplied by the periodic modulation

$$B_w(t) = \sqrt{2qI_c(t)}$$

for the collector current shot noise of the transistor, and then the contribution of this noise source to c_w in (65) is calculated as the time-average of the square of this periodic quantity. Fig. 3(c)

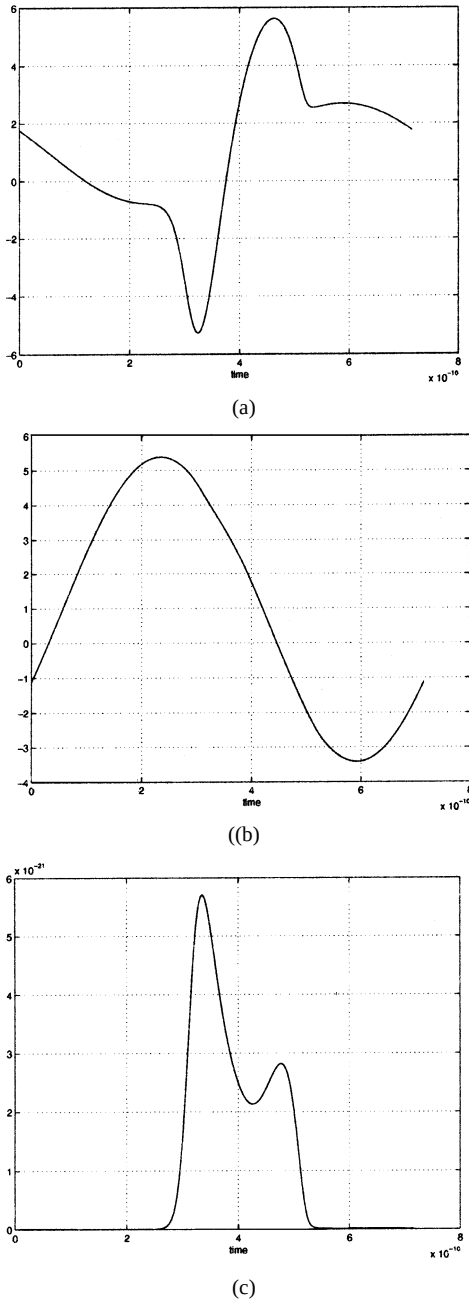


Fig. 3. Colpitts oscillator phase noise analysis.

shows $v_1^T(t)B_w B_w^T v_1(t)$ for all of the white-noise sources, i.e., the collector current shot noise, the base current shot noise, and the thermal noise sources of the collector, base and emitter resistances. c_w is the time-average of this quantity.

The waveform $v_1(t)$ for *base-emitter* voltage is used in calculating the contribution of the $1/f$ and the burst noise source connected between the base and the emitter of the transistor in its noise model [6]. As observed in Fig. 3(a), this waveform has a *nonzero* time-average. Any colored-noise source connected between the base and the emitter will have an influence on the spectrum of the oscillator due to phase noise.

Fig. 4 shows the computed single-sideband phase noise spectrum (that can be measured by a specialized equipment) for an oscillator which has both white and flicker noise sources. This spectrum is computed using the expression in (69) for $f \gg 0$.

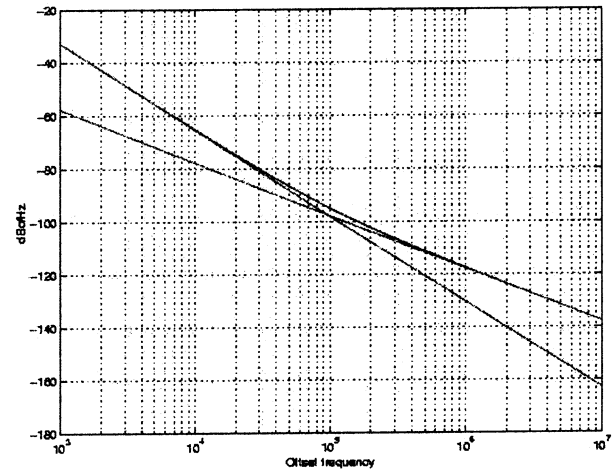


Fig. 4. Single-sideband phase noise spectrum with white and $1/f$ noise sources.

IX. CONCLUSION

We have presented the theory and practical characterization of phase noise in oscillators due to *colored*, as opposed to white, noise sources, based on a *nonlocal Fokker-Planck equation* that we derived and solved. The results of this paper together with the numerical methods described in [2] have been implemented in a proprietary circuit simulator. It has been used by oscillator designers and proved to be a very useful tool in the search for novel oscillator circuit architectures with improved phase noise.

APPENDIX

Proof of Theorem IV.1

Let $q_\alpha(\eta, t)$ be the time-varying *conditional* probability density function of $\alpha(t)$ *given* $b(t)$:

$$q_\alpha(\eta, t) = \frac{\partial P_b(\alpha(t) \leq \eta | b(t))}{\partial \eta}, \quad t \geq 0 \quad (70)$$

where $P_b(\cdot | b(t))$ denotes the *conditional* probability measure given $b(t)$. Let $E_b[\cdot | b(t)]$ denote the *conditional expectation* operator associated with the conditional probability measure $P_b(\cdot | b(t))$. Let $h(z)$ be an arbitrary but smooth function. From (10) we obtain

$$\begin{aligned} \frac{dh(\alpha(t))}{dt} &= \left. \frac{dh(z)}{dz} \right|_{z=\alpha(t)}, & \frac{d\alpha(t)}{dt} \\ &= \left. \frac{dh(z)}{dz} \right|_{z=\alpha(t)}, & v(t + \alpha(t))b(t). \end{aligned} \quad (71)$$

Now, we apply the conditional expectation operator $E_b[\cdot | b(t)]$ to both sides of the above equation

$$\begin{aligned} E_b \left[\left. \frac{dh(\alpha(t))}{dt} \right| b(t) \right] \\ = E_b \left[\left. \frac{dh(z)}{dz} \right|_{z=\alpha(t)} v(t + \alpha(t))b(t) \right] b(t) \end{aligned} \quad (72)$$

which can be rewritten using the conditional PDF $q_\alpha(\eta, t)$ of $\alpha(t)$ as follows:

$$\int_{-\infty}^{\infty} \frac{d}{dt} h(\eta) q_\alpha(\eta, t) d\eta = \int_{-\infty}^{\infty} \frac{dh(\eta)}{d\eta} v(t+\eta) b(t) q_\alpha(\eta, t) d\eta. \quad (73)$$

We use integration by parts to rewrite the integral on the right-hand side to obtain

$$\int_{-\infty}^{\infty} \frac{\partial}{\partial t} h(\eta) q_\alpha(\eta, t) d\eta = - \int_{-\infty}^{\infty} h(\eta) \frac{\partial}{\partial \eta} [v(t+\eta) b(t) q_\alpha(\eta, t)] d\eta \quad (74)$$

where we assumed that the conditional PDF $q_\alpha(\eta, t)$ vanishes at $\eta = \pm\infty$ and

$$\frac{\partial}{\partial t} q_\alpha(\eta, t) = - \frac{\partial}{\partial \eta} [v(t+\eta) b(t) q_\alpha(\eta, t)] \quad (75)$$

follows immediately, since (74) is satisfied for an arbitrary function $h(\eta)$. Now, we use techniques from [10] to proceed further. We rewrite (75) as follows:

$$\frac{\partial}{\partial t} q_\alpha(\eta, t) = L_\eta(t) b(t) q_\alpha(\eta, t) \quad (76)$$

with the operator

$$L_\eta(t) = - \frac{\partial}{\partial \eta} v(t+\eta). \quad (77)$$

Equation (76) has the formal solution

$$q_\alpha(\eta, t) = \mathcal{A} \exp \left[\int_0^t L_\eta(\tau) b(\tau) d\tau \right] q_\alpha(\eta, 0). \quad (78)$$

Next, we will take the expectation of both sides of the above solution over all realizations of the noise source $b(t)$

$$\mathbb{E}[q_\alpha(\eta, t)] = \mathcal{A} \mathbb{E} \left[\exp \left\{ \int_0^t L_\eta(\tau) b(\tau) d\tau \right\} q_\alpha(\eta, 0) \right]. \quad (79)$$

From the definitions of $p_\alpha(\eta, t)$ and $q_\alpha(\eta, t)$, in (11) and (70), we have

$$p_\alpha(\eta, t) = \mathbb{E}[q_\alpha(\eta, t)]$$

where the expectation is over all realizations of $b(t)$. We would like to obtain a closed equation for the marginal unconditional PDF $p_\alpha(\eta, t)$ of $\alpha(t)$. We can accomplish this if we can express the expectation on the right-hand-side of (79) in terms of $p_\alpha(\eta, t)$. In the general case, this is not possible, because $\exp[\int_0^t L_\eta(\tau) b(\tau) d\tau]$ and $q_\alpha(\eta, 0)$ are *correlated*. However, if we restrict ourselves to the case when $p_\alpha(\eta, 0) = q_\alpha(\eta, 0) = \delta(\eta - \alpha_0)$, then, we obtain

$$p_\alpha(\eta, t) = \mathcal{A} \mathbb{E} \left[\exp \left\{ \int_0^t L_\eta(\tau) b(\tau) d\tau \right\} p_\alpha(\eta, 0) \right]. \quad (80)$$

The expectation in (80) can be evaluated exactly, using the fact that $b(t)$ hence $\int_0^t L_\eta(\tau) b(\tau) d\tau$ is a Gaussian process

$$\begin{aligned} p_\alpha(\eta, t) &= \mathcal{A} \exp \left[\frac{1}{2} \int_0^t \int_0^t L_\eta(t_1) L_\eta(t_2) R_N(t_1 - t_2) dt_1 dt_2 \right] p_\alpha(\eta, 0) \\ &= \mathcal{A} \exp \left[\frac{1}{2} \int_0^t \int_0^t L_\eta(t_1) L_\eta(t_2) R_N(t_1 - t_2) dt_1 dt_2 \right] p_\alpha(\eta, 0) \end{aligned} \quad (81)$$

follows using the form of the characteristic function for a Gaussian random variable. $R_N(\tau)$ is the autocorrelation function of $b(t)$ given in (7). If we take the time derivative of both sides of the above, we obtain

$$\frac{\partial}{\partial t} p_\alpha(\eta, t) = L_\eta(t) \int_0^t L_\eta(\tau) R_N(t - \tau) p_\alpha(\eta, t) d\tau. \quad (82)$$

We substitute the operator $L_\eta(t)$ from (77) into the above

$$\begin{aligned} \frac{\partial}{\partial t} p_\alpha(\eta, t) &= \frac{\partial}{\partial \eta} \\ &\cdot \left\{ v(t+\eta) \int_0^t \frac{\partial}{\partial \eta} [v(\tau+\eta) R_N(t-\tau) p_\alpha(\eta, t)] d\tau \right\}. \end{aligned} \quad (83)$$

Equation (14) follows from the above equation through the chain rule for derivatives. ■

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